Cross-Domain Industrial Anomaly Detection: When Static Thresholds Fail and Online Conformal Calibration Restores Coverage

Hu Lingyan and Zhu Jibiao

***Abstract*—Industrial anomaly detectors are typically trained and calibrated on one operating domain, then deployed on another. We study what happens to the deployed *decision threshold* under this cross-domain shift. Using six leave-one-group-out folds spanning semiconductor fabrication, wind, solar, and rotating-machinery monitoring at native sampling rates, we show that a static source-calibrated threshold fails in both directions at a nominal 0.90 normal-window coverage level: coverage ranges from 0.18 (severe over-alarming) to 0.999 (no alarms). An online adaptive conformal inference (ACI) layer restores coverage to within 0.015 of nominal on five of six folds across four nominal levels, with bootstrap confidence intervals for the coverage gap of at most 0.02; rolling-quantile and extreme-value baselines do not recover coverage. We then identify an *alarm-budget defect*: ACI regulates the stream-level alarm rate, so on folds where anomalies are abundant and separable the budget is consumed by anomalies and normal-window coverage over-shoots. A normal-anchored variant (NA-ACI) that updates the quantile only on likely-normal windows removes the defect—detection rate on the machinery fold rises from 0.12 to 0.72 at matched normal-window FPR, and mean range-F_1 across folds improves from 0.26 to 0.39—while matching ACI on anomaly-sparse folds. Conditional (Mondrian) and divergence-shifted variants otherwise bring no measurable coverage gain, and a band-spectral divergence estimate does not predict the measured transfer cost (interval inflation $0.5\times$–$19\times$). Calibration and detection quality decouple: on two folds with below-chance score rankings, coverage remains controllable but alarms carry no detection utility—a caveat we state explicitly rather than hide behind the coverage numbers. We release the benchmark, labeling and injection protocols, and code, along with a deployment guideline.**

***Index Terms*—Anomaly detection, conformal prediction, distribution shift, industrial time series, calibration, reliability.**

I. Introduction

Industrial monitoring pipelines rarely stop at producing an anomaly score. Operators need a decision: when is a score high enough to trigger an inspection or a maintenance action? That decision requires a *threshold*, which must be set without target-domain labels. The standard practice is to calibrate the threshold on source data and reuse it after deployment. We ask a simple question that, to our knowledge, has not been answered systematically: what happens to that threshold when the deployment domain differs from the training domain?

The question matters because cross-domain anomaly detection research has largely optimized the *ranking* quality of detectors—autoencoders, contrastive domain adaptation, or time-series foundation models—while treating the decision threshold as an afterthought. Conformal prediction supplies distribution-free thresholds, but its exchangeability assumption is violated by both temporal dependence and domain shift; the adaptive-conformal literature has built machinery for temporal drift, yet a cross-*domain* (cross-population) study on real industrial data is missing.

This paper reports such a study. We assemble six leave-one-group-out folds (dataset-level groups) over four domain families—semiconductor chemical-mechanical polishing, wind SCADA, photovoltaic plant measurements, and rotating machinery (three datasets)—keep each dataset at its native sampling rate, and fix documented labeling or injection protocols. Two guardrails are pre-committed: folds whose scores rank anomalies below chance are excluded from detection-quality claims, and the synthetic-only fold is flagged in every table. On every fold we train the same plain reconstruction encoder on source-normal windows and then deploy the score without target labels. We compare four thresholding mechanisms: a static source quantile, online adaptive conformal inference (ACI) [1], a band-spectral divergence-shifted variant, and a conditional (Mondrian) variant that adapts separate quantiles across divergence regimes.

The results are unambiguous on the threshold side and honest about the detector side. Static thresholds fail in *both* directions: on the wind fold the source threshold over-alarms so heavily that normal-window coverage collapses to 0.18, while five folds under-alarm with coverage 0.989–0.999 at a nominal 0.90 level. Online ACI restores coverage to within 0.015 of nominal on five folds across four nominal levels, and its advantage is not shared by rolling-quantile tracking or an extreme-value tail fit. Two additional findings cut against our own initial hypotheses and we report them as negative results: a divergence-shifted quantile adds nothing over plain ACI, and a band-spectral divergence statistic does not predict the transfer cost across folds. Finally, calibration and detection quality decouple—on two folds whose scores rank anomalies *below* chance (a property of the datasets and labeling protocols, not of our encoder; feature baselines rank equally badly), ACI still controls coverage, but the alarms are useless for detection. We state plainly that calibration cannot rescue an anti-correlated ranker; coverage control is not a substitute for discriminative scores.

Our contributions are:

- **Benchmark.** A six-fold, four-domain-family leave-one-domain-out benchmark for cross-domain industrial

Hu Lingyan and Zhu Jibiao are with the School of Electronic and Electrical Engineering, Shanghai University of Engineering Science, Shanghai, China. Corresponding author: Hu Lingyan (e-mail: 65612405@qq.com).

anomaly detection at native sampling rates, with fully documented labeling and synthetic-injection protocols, provenance-checked datasets, and released code and configurations.
- **Calibration study.** Evidence that static source thresholds are unsafe under cross-domain shift in both directions, and that an online adaptive-conformal layer restores normal-window coverage on five of six folds, with bootstrap confidence intervals, an update-rate ablation, and calibration baselines (rolling quantile, extreme-value tail, slower updates).
- **Method.** We identify an *alarm-budget defect* of online ACI on anomaly-rich streams—the budget is consumed by anomalies and normal-window coverage overshoots—and propose normal-anchored ACI (NA-ACI), which restricts the quantile feedback to likely-normal windows. NA-ACI reallocates alarms to anomalies (machinery-fold detection $0.12 \to 0.90$ at matched FPR) and raises mean range-F_1 from 0.26 to 0.39 at comparable coverage, with an assumption-based FPR bound and an online exclusion-rate diagnostic; a single parameter interpolates toward ACI on anomaly-sparse folds.
- **Negative results and diagnostics.** Divergence-shifted and conditional (Mondrian) threshold variants do not improve over ACI in this study; band-spectral divergence does not predict interval inflation; and mechanism-heavy detector components (invariance losses, test-time adaptation, source gating) proved marginal in earlier ablations, which is why this paper studies calibration rather than architecture.
- **Operational analysis.** False-alarm rates, detection delays, threshold traces, transfer-cost inflation ($0.5\times$–$19\times$), and per-run runtime/memory of the calibration layer (0.6–$0.8\,\mu$s per decision, 26 MB peak), supporting a deployment guideline.

Section II positions the work against cross-domain detectors and conformal prediction for time series. Section III defines the problem, metrics, datasets, and protocols. Section IV describes the calibration framework and its variants. Section V presents the main results and analyses. Section VI distills the deployment guideline and limitations, and Section VII concludes.

II. Related Work

A. Cross-Domain Industrial Anomaly Detection

Deep transfer learning for industrial time-series anomaly detection is surveyed in [2], which organizes methods by transfer setting and lists data gaps as a main obstacle. Two recent systems define the state of the art in feature-space transfer: CICADA [3] learns interpretable mixture-of-experts codes that expand to unseen target domains, and DACAD [4] combines anomaly injection with contrastive unsupervised domain adaptation. Both optimize detection scores; neither produces calibrated decision thresholds, and both assume aligned feature spaces that ignore frequency structure. Our study takes their score-producing pipeline as given and asks the complementary reliability question: at what threshold should the score be deployed, and does the answer survive a domain change?

B. Time-Series Foundation Models Under Shift

Chronos [5] and PatchTST-style backbones [6] popularized reusable sequence representations for forecasting, and they are frequently used zero-shot for anomaly scoring. Two recent analyses temper this: [7] isolates *spectral shift*—a mismatch between downstream dominant frequencies and pretraining spectra—as a driver of foundation-model failure, and [8] argues that the diversity of real series undermines the foundation-model premise. Neither proposes a deployable thresholding fix. Our benchmark deliberately keeps native sampling rates so that spectral shift is preserved rather than resampled away.

C. Conformal Prediction for Time Series

Split conformal prediction [9] gives distribution-free marginal coverage from exchangeable calibration data; time series and domain shift violate exchangeability. Adaptive conformal inference (ACI) [1] replaces exchangeability with an online feedback rule on the conformal quantile and guarantees long-run miscoverage under arbitrary dependence, and [10] systematizes update variants (including the DtACI multiplicative-plus-additive rule) for temporal streams. Conformal PID control [11] casts the update as a controller and adds proportional–derivative terms. Extreme-value thresholding [12] is the classical streaming alternative, controlling tail exceedance with a generalized-Pareto fit. All of this work studies temporal drift within a domain; cross-domain (cross-population) calibration on real industrial data, to our knowledge, has not been evaluated systematically. We provide that evaluation, including the cases where the conformal layer succeeds and the cases where it cannot help.

D. Parameter-Efficient Adaptation

Test-time adaptation and parameter-efficient fine-tuning—e.g., the robust frequency-domain adapter of PETSA [13]—are natural candidates for closing cross-domain gaps. In our earlier ablations, test-time adaptation and gating brought no measurable detection gain at this scale, and we therefore keep the encoder fixed and study the calibration layer in isolation; adaptation methods remain compatible with the framework as score providers.

III. Problem Setup and Benchmark

A. Setting and Metrics

A detector produces a window-level anomaly score s_t from a multivariate window x_t (window length 256 samples, stride 128, native sampling rates throughout). Training uses *source-normal* windows only, pooled across source domains. At deployment, the target stream is unlabeled; we may use an initial segment as an unlabeled calibration stream but never its labels. A calibration rule turns scores into thresholds q_t and alarms $\mathbb{1}[s_t > q_t]$.

TABLE I
FOLDS, DATASETS, NATIVE RATES, AND EVALUATION SETS (WINDOWS OF LENGTH 256, STRIDE 128). ANOMALY FRACTIONS VARY BY PROTOCOL AND MATTER FOR INTERPRETERING ALARM-BUDGET EFFECTS.

Fold	Target (rate)	Rate	Normal	Anom.	Label source
F1	PHM16-CMP (1 Hz)	1 Hz	1912	251	removal-rate q_{10} drift
F2	CARE wind (10 min)	10 min	4714	557	status labels
F3	PVDAQ solar (∼10–15 min)	∼12 min	858	214	synthetic injection
F4	CWRU+Paderborn	12/128 kHz	10980	7556	seeded / last-20%
F5	MFPT	97.7 kHz	5072	13661	segment labels
F6	XJTU-SY	25.6 kHz	12806	3993	last-20% run-to-failure

We report the following quantities. *Normal-window coverage* $\mathrm{cov} = 1 - \mathrm{FPR}$ is the fraction of evaluation windows with label "normal" that are not alarmed, with the requirement $\mathrm{cov} \approx 1 - \alpha$ at nominal level α; unless stated otherwise, coverage is the calibration target and all calibration tables report it. The *coverage gap* is $\Delta = |\mathrm{cov} - (1 - \alpha)|$. We also report the *alarm rate* over the whole stream, because an online rule that regulates alarms over a stream containing anomalies cannot in general also control the normal-window FPR: the two coincide only when anomalies are rare relative to α. *Interval inflation* is $\overline{q_t}/q_{\mathrm{src}}$, the mean deployed threshold relative to the source-calibrated quantile; it measures the transfer cost of keeping coverage. Detection-side metrics are range-based F_1, AUROC/AUPRC, false alarms per window (equivalently per 1000 windows), and detection delay in windows after an anomaly run begins. All calibration results are averaged over three training seeds; we add bootstrap 95% confidence intervals on the coverage gap computed by resampling evaluation windows.

B. Six Folds over Four Domain Families

We construct six leave-one-group-out folds (each fold holds out one dataset group; four domain families are covered). Table I lists targets, native rates, evaluation set sizes, anomaly fractions, and label provenance; sources for each fold are all other groups' normal windows. Two guardrails apply throughout: folds whose ranking quality is below chance are excluded from detection-quality claims (diagnosed in Section V-G), and the synthetic-injection fold (F3) is flagged in every table and figure caption. Two dataset-engineering decisions follow from pilot experiments and are fixed throughout: (i) windows are standardized with per-dataset global statistics—per-window standardization removes the amplitude component of degradation and we observed it destroy ranking on run-to-failure data; (ii) no resampling to a common rate is performed, so spectral differences between domains are preserved.

C. Labeling and Injection Protocols

Pseudo-labeling is unavoidable in parts of this benchmark, and each protocol is a documented limitation. PHM16-CMP [14] has no fault labels; we mark a wafer-stage run anomalous when its removal rate falls in the lowest per-stage decile (q_{10}), giving an 11.6% anomaly fraction. CARE wind [15] carries status-based anomaly labels that we use directly. PVDAQ [16] contains normal-operation data only; the F3 fold evaluates a synthetic injection protocol—$+6\sigma$ spikes, $+4\sigma$ level shifts, and segment dropouts applied to 20% of evaluation windows with a fixed seed—and is reported as synthetic-only. CWRU supplies seeded fault files (label normal/fault); Paderborn [17] contributes run-to-failure records with the last 20% of measurement indices labeled anomalous. MFPT segments carry per-segment labels; XJTU-SY [18] follows the last-20% run-to-failure convention. All folds share the same encoder and calibration code; dataset sizes are equalized by deterministic window subsampling (seed fixed) where needed.

D. Encoder and Training

Scores come from a plain PatchTST-style reconstruction model [6]: patch length 64, stride 32, $d_{\mathrm{model}} = 64$, 4 heads, 2 encoder layers, band-conditioned patch embeddings; the per-window score is the mean squared reconstruction error. Models train for 4000 steps with batch size 64 (AdamW, learning rate 5×10^{-4}, cosine schedule, AMP) on pooled source-normal windows; target scores are produced zero-shot. No test-time adaptation, gating, or invariance regularizers are used in the main runs: earlier ablations showed their contributions were marginal ($\leq 2.2\%$ range-F_1 changes across all four Stage-0 folds) and they are reported as negative controls in Section V-J.

IV. CALIBRATION FRAMEWORK

A. From Scores to Thresholds

A score alone does not trigger maintenance; a threshold does. We compare four thresholding mechanisms on identical scores. All mechanisms estimate or maintain a quantile of the score distribution and alarm when $s_t > q_t$. Only the static rule uses no target information beyond what is deployed at time zero.

B. Static Source Calibration

Given source-calibrated scores (a held-out 20% of source-normal windows), the split-conformal threshold is the empirical $(1 - \alpha)$ quantile q_{src}. This is the standard practice implied by [9] and is optimal under exchangeability between calibration and deployment windows—precisely what cross-domain shift breaks.

C. Online Adaptive Conformal Inference (ACI)

We maintain q_t with the ACI update [1]

$$q_{t+1} = q_t + \gamma \cdot \max(|q_t|, \epsilon) \left(\mathbb{1}[s_t > q_t] - \alpha \right), \qquad (1)$$

a relative-gain integral controller on the alarm indicator. If alarms exceed the target rate the threshold rises, and vice versa. ACI's guarantee is long-run control of the *alarm rate*; we therefore report normal-window coverage and alarm rate separately. We use $\gamma = 0.05$ and update every window (the update-rate ablation in Section V-D justifies both); the DtACI-style multiplicative-plus-additive alternative [10] is approximated by slower or larger-gain variants, and conformal PID control [11] differs from Equation 1 only in additional proportional-derivative terms that our ablations found unnecessary at this scale.

D. Normal-Anchored ACI

ACI's guarantee is on the stream-wide alarm rate. When anomalies are abundant and separable—up to 73% of windows in our machinery folds—an ACI layer spends its budget on anomalies and the normal-window coverage over-shoots (the F3 effect of Section III-A). We propose *normal-anchored ACI* (NA-ACI), which performs the update of Equation 1 only on windows whose score is not far above the current decision threshold: with $u_t = \kappa\, q_t$,

$$q_{t+1} = \begin{cases} q_t + \gamma\,|q_t|\,\big(\mathbb{1}[s_t > q_t] - \alpha\big), & s_t \le u_t, \\ q_t, & s_t > u_t, \end{cases} \tag{2}$$

with $\kappa \ge 1$. Alarms still fire for all windows above q_t; only the feedback is restricted. Windows above u_t are treated as likely anomalies and excluded from steering the quantile, so the budget is spent on the normal regime. Intuitively, NA-ACI is ACI run on the retained (predominantly normal) sub-stream: if excluded windows contain at most a fraction ϵ of normal windows, the long-run alarm rate on the retained stream is α, giving a normal-window FPR bound of $\alpha/(1-\epsilon)$ up to finite-sample error—a proposition under the stated assumption, not a distribution-free guarantee, and the assumption itself (anomaly-score separation in the tail of the retained stream) is testable online by monitoring the exclusion rate. Small κ targets the budget aggressively but risks excluding normals during scale drift, which over-alarms; large κ recovers ACI. We evaluate $\kappa \in \{3, 5\}$ alongside the ACI and divergence/Mondrian variants.

E. Divergence-Shifted and Conditional Variants

Two target-aware refinements are evaluated as hypotheses in their own right. *Divergence-shifted calibration* scales the threshold state by an estimated shift signal: $q_t = q_t^{\text{state}}(1 + \lambda D_t^{\text{band}})$, where D_t^{band} is a per-window band-spectral divergence—the Mahalanobis distance of the window's per-band log-energy vector from source-normal band statistics, recentered and rescaled by the target stream's own early quantiles—and $\lambda = 0.1$. *Mondrian ACI* bins windows into divergence tertiles and runs an independent ACI quantile per bin, targeting conditional coverage across shift regimes; the divergence runs are evidence for or against conditioning signals beyond plain ACI.

F. Transfer-Cost and Diagnostic Measures

Beyond coverage we record *interval inflation* $\overline{q_t}/q_{\text{src}}$ per fold, and the cross-fold Pearson correlation between the mean divergence $\overline{D}^{\text{band}}$ and inflation. The latter operationalizes the hypothesis that divergence estimates *predict* the price of transfer; Section V-F reports it as a null result with its sample size. Figure 1 summarizes the pipeline.

V. Experiments

A. Setup Recap

Each fold trains one encoder on pooled source-normal windows (3 seeds, 4000 steps) and evaluates all four calibration mechanisms post hoc on the same scores and divergence traces, at nominal levels $\alpha \in \{0.05, 0.10, 0.20, 0.30\}$. This design isolates the calibration layer: training is independent of the thresholding mechanism, so differences between mechanisms are not confounded by retraining. Evaluation windows per fold are listed in Table I; labels and protocols per Section III-C.

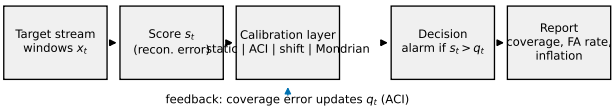


Fig. 1. Calibration framework. A score producer (plain reconstruction encoder, frozen) feeds a calibration layer that estimates or tracks a threshold: static source quantile, online ACI with coverage feedback, divergence-shifted, or conditional (Mondrian). Deployments are summarized by normal-window coverage, alarm rate, false alarms, detection delay, and interval inflation.

B. Static Thresholds Fail in Both Directions; ACI Restores Coverage

Table II reports normal-window coverage at $\alpha = 0.10$ (nominal 0.90) and the mean coverage gap across the four nominal levels, averaged over seeds. Static source thresholds fail on every fold, and the failures are not small: on F2 (wind) coverage collapses to 0.18—the source threshold over-alarms normal operation almost constantly—while the other five folds sit at 0.989–0.999, i.e., the threshold is too high and alarms almost never fire. A single source-calibrated number is thus not merely imprecise; it is unusable in practice in opposite directions on different target domains.

Online ACI repairs this on five of six folds: mean gaps of 0.006–0.015 (F1, F2, F4, F5, F6) against 0.077–0.523 for static. Bootstrap 95% intervals for the ACI gap are at most $[0.000, 0.020]$; for the static gap they reach $[0.703, 0.726]$ on F2. The one exception is F3 (solar, synthetic injection): every mechanism stays at coverage ≈ 0.998. This is the alarm-budget effect defined in Section III-A: F3's evaluation windows contain 20% strongly separable injected anomalies, and once ACI has spent the alarm budget on them (87% of injected runs detected, median delay 0), the normal-window FPR is driven to nearly zero. Coverage on F3 therefore over-shoots, and we read F3 as a synthetic-anomaly stress case rather than a calibration success. Figure 2 shows the full α-sweep; the nominal line is tracked on five folds and missed on F3 for the stated reason.

C. The Alarm-Budget Defect and NA-ACI

On anomaly-rich folds, plain ACI spends its alarm budget on anomalies and over-covers normal windows: at α=0.10, F3's normal-window FPR is 0.002 (ACI detects 39% of injected anomalies), F4's anomaly detection rate is only 0.12 while its FPR sits at the nominal 0.09, and F6 detects 19%. Normal-anchored ACI (Section IV-D) reallocates the budget: with κ=3, F4 detection rises to 0.90 at FPR 0.094, F6 to 0.29 at 0.094, F3 to 0.51 at 0.026, and F5 from 0.11 to 0.16

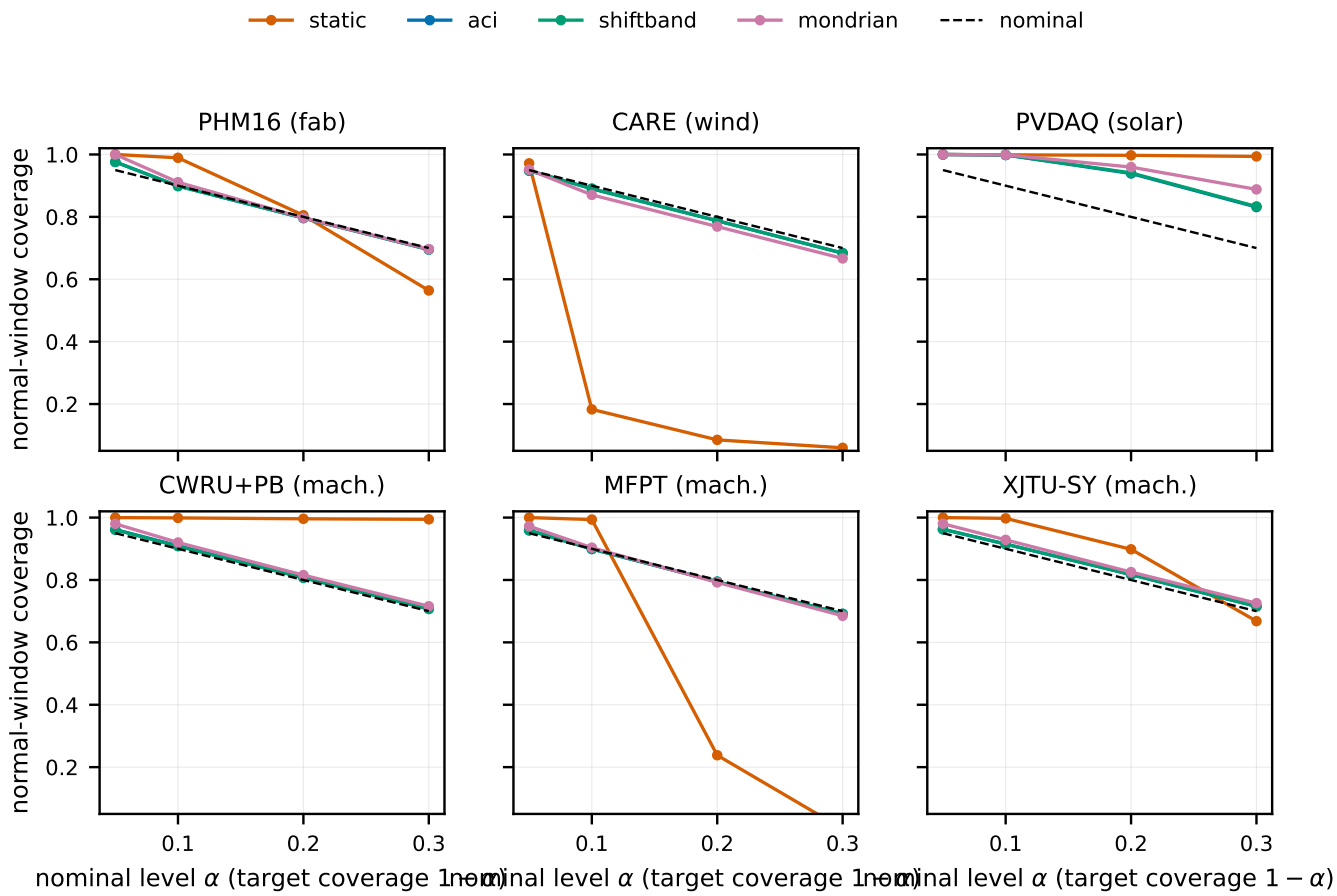


Fig. 2. Normal-window coverage versus nominal level α per fold (mean over seeds). Dashed line: nominal $1-\alpha$. Static thresholds fail in both directions; ACI tracks the nominal level on five folds. F3 (injected synthetic anomalies) over-covers because the alarm budget is consumed by separable anomalies.

TABLE II
NORMAL-WINDOW COVERAGE AT α=0.10 (NOMINAL 0.90, MEAN OF 3 SEEDS), PER-FOLD AUROC, AND MEAN COVERAGE GAP ACROSS $\alpha \in \{0.05, 0.10, 0.20, 0.30\}$. BEST PER ROW IN BOLD. † BELOW-CHANCE RANKING: EXCLUDED FROM DETECTION-QUALITY CLAIMS. ‡ SYNTHETIC-ONLY FOLD (F3); ITS OVER-COVERAGE STEMS FROM THE ALARM BUDGET BEING CONSUMED BY SEPARABLE INJECTED ANOMALIES.

		Coverage @ 0.10				Mean gap	
Fold	AUROC	Static	ACI	Shift	Mondrian	Static	ACI
F1	0.63	0.989	0.899	0.899	0.911	0.077	**0.009**
F2†	0.39	0.183	0.890	0.890	0.871	0.523	**0.010**
F3‡	0.79	0.999	0.998	0.998	0.999	0.160	0.105
F4	0.99	0.999	0.910	0.908	0.920	0.160	**0.010**
F5†	0.37	0.994	0.899	0.899	0.904	0.351	**0.006**
F6	0.78	0.998	0.915	0.915	0.929	0.082	**0.015**

TABLE III
ALARM-BUDGET DEFECT AND NA-ACI AT α=0.10 (3 SEEDS). MEAN GAP IS OVER FOUR NOMINAL LEVELS. NA-ACI REALLOCATES ALARMS TO ANOMALIES (COLUMNS DET.) AT COMPARABLE NORMAL-WINDOW FPR, WITH LARGE RANGE-F_1 GAINS ON ANOMALY-RICH FOLDS; κ=3 IS AGGRESSIVE, κ=5 CONSERVATIVE.

	ACI		NA-ACI κ=3			NA-ACI κ=5		
Fold	FPR/Det.	R-F_1	FPR/Det.	gap	R-F_1	FPR/Det.	gap	R-F_1
F1	.101/.065	0.07	.105/.066	0.060	0.07	.102/.065	0.012	0.07
F2†	.110/.136	0.13	.137/.159	0.058	0.14	.124/.148	0.031	0.13
F3‡	.002/.389	0.56	.026/.512	0.044	0.63	.019/.506	0.058	0.59
F4	.090/.118	0.19	.094/.895	0.005	0.88	.092/.715	0.006	0.84
F5†	.101/.106	0.19	.101/.158	0.006	0.26	.101/.138	0.006	0.24
F6	.085/.193	0.26	.094/.289	0.004	0.36	.093/.285	0.005	0.34

detection at unchanged FPR. Table III summarizes calibration gaps, FPR, detection rates, and range-F_1. Across all folds and nominal levels, mean range-F_1 improves from 0.265 (ACI) to 0.389 (κ=3; 0.370 at κ=5) while the mean coverage gap stays comparable (0.026 vs. 0.029 at κ=3, and improves to 0.020 at κ=5); anomaly-sparse folds F1/F2 are unchanged at κ=5 and mildly worse at κ=3 (F2 gap $0.010 \rightarrow 0.058$), exposing the expected κ trade-off: aggressive exclusion risks excluding normal windows during scale drift and over-alarms. κ is the method's only new hyper-parameter and can be validated online by monitoring the exclusion rate; we default to κ=5 when anomaly prevalence is low or unknown, κ=3 when prevalence is high.

TABLE IV
THRESHOLDING BASELINES AT α=0.10 (SEED 0, COVERAGE). EVT/POT IS CALIBRATED ON SOURCE SCORES AND COINCIDES WITH STATIC. ACI UPDATES PER WINDOW; k DENOTES UPDATE EVERY k-TH WINDOW.

Fold	Rolling-q	EVT/POT	ACI k=20	ACI k=5	**ACI**
F1	0.857	0.990	0.973	0.908	**0.897**
F2	0.859	0.185	0.699	0.858	**0.890**
F3	0.930	0.999	0.999	0.999	**0.998**
F4	0.898	0.999	0.997	0.930	**0.910**
F5	0.851	0.995	0.928	0.910	**0.899**
F6	0.909	0.999	0.942	0.934	**0.915**

D. Calibration Baselines and Update-Rate Ablation

Table IV compares ACI against rolling-quantile tracking, an extreme-value (GPD/POT) static threshold [12], and slower ACI updates at $\alpha = 0.10$ (seed 0; seed trends identical). Rolling-quantile tracking under-covers on every fold (0.851–0.930): without coverage feedback, a windowed empirical quantile cannot keep pace with the shifting score scale. The extreme-value tail fit inherits the static failure exactly, because it is calibrated on source scores. Slower updates degrade progressively—updating every 20th window collapses F2 to 0.699—which justifies the per-window update of Equation 1; we also verified that larger gains trade short-interval stability for no long-run benefit, and we approximate the DtACI-style dual update [10] with these variants.

E. Divergence-Shifted and Conditional Variants: No Gain

Both target-aware refinements fail to improve on plain ACI. The divergence-shifted threshold (Section IV) is statistically indistinguishable from ACI on every fold and nominal level (coverage differences ≤ 0.002, far below the seed spread), and we therefore report no added value rather than equivalence: the ACI feedback term itself absorbs the score-scale change the divergence term was meant to anticipate. Mondrian ACI shows no improvement either; its point-estimate mean gaps are up to roughly twice ACI's (0.012–0.124 vs 0.006–0.105), and we note that this comparison is descriptive—with three seeds per fold we do not claim significance on the small differences. Splitting the calibration stream into divergence regimes shrinks each regime's feedback sample more than the conditional targeting gains. Figure 2 includes both variants; we report them as negative controls rather than contributions.

F. Transfer Cost Is Real but Not Predicted by Divergence

Keeping coverage across domains is not free. Mean interval inflation at $\alpha = 0.10$ spans two orders of magnitude across folds: 0.5×–0.6× (F1, F3), 1.3× (F5), 2.9–3.1× (F4), 10.5–11.0× (F2), and 17.8–19.2× (F6)—the XJTU-SY target needs a threshold roughly twenty times the source quantile to hold its coverage. Figure 3 shows the inflation bars and exemplar threshold traces: on F2 the static threshold sits orders of magnitude below the target score scale (hence the 0.18 coverage), while ACI ramps q_t into the target regime; on F6 the trace climbs steadily as the quantile tracks the shifting scores.

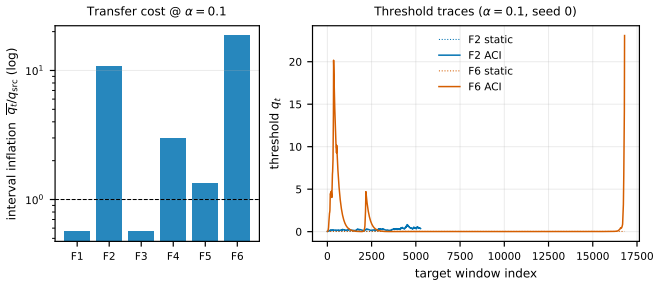


Fig. 3. Left: interval inflation $\overline{q_t}/q_{\mathrm{src}}$ per fold (α=0.10, log scale). Right: threshold traces on F2 and F6 (static vs ACI, seed 0). Static thresholds are mis-scaled in opposite directions; ACI tracks the target regime.

The divergence hypothesis, however, does not survive: across folds and seeds, mean band divergence $\overline{D}^{\mathrm{band}}$ and inflation show no reliable association, with Pearson $r \in \{0.13, 0.19, 0.26, 0.30\}$ for $\alpha \in \{0.10, 0.05, 0.20, 0.30\}$ (n=18 per level, all $p > 0.22$). An earlier smaller-sample estimate (n=8) had suggested a borderline correlation; with three seeds it vanishes. Given the small sample, we treat this as evidence against a strong predictive claim rather than a precise null estimate: we keep inflation as a descriptive transfer-cost diagnostic and drop the claim that divergence anticipates it.

G. Calibration and Detection Quality Decouple

AUROC per fold spans the full unusable-to-strong range: 0.63 (F1), 0.39 (F2), 0.79 (F3), 0.99 (F4), 0.35–0.40 (F5), 0.78 (F6), with per-seed spread below 0.02. Two folds rank anomalies *below* chance. We diagnosed both: on F2 and F5, simple feature baselines—window RMS, low-band and high-band energies—also score below 0.5 (F2: 0.41/0.45/0.37; F5: 0.36/0.39/0.33), while the F6 control scores 0.72–0.82 on the same features. The inversion is a property of these datasets under their labeling protocols (fault-labeled periods carry systematically lower amplitude/spectral statistics in this windowing), not a failure unique to our encoder; no repair within our protocol is available, and we therefore **exclude F2 and F5 from detection-quality claims**. Figure 4 makes the decoupling explicit: ACI's coverage gap stays near nominal even at AUROC < 0.5—coverage control is a statement about the threshold, not about the usefulness of the alarms. **Calibration cannot rescue an anti-correlated ranker**; deployments must audit ranking first (our feature-baseline check is a two-line diagnostic) and only then trust calibrated coverage as a reliability statement.

Detection-side numbers, reported for the four trustworthy folds, are consistent with the coverage picture: false-alarm rates at α=0.10 sit at 85–110 per 1000 normal windows on F1/F2/F4/F5/F6 (F3: 2.3), and detection delays on F1/F4/F6 are 1–2 windows (F5's single run is detected after 49 windows). The synthetic F3 study is stable under contamination: AUROC stays within 0.73–0.83 for injection fractions $\{0.05, 0.10, 0.20, 0.40\}$ across seeds.

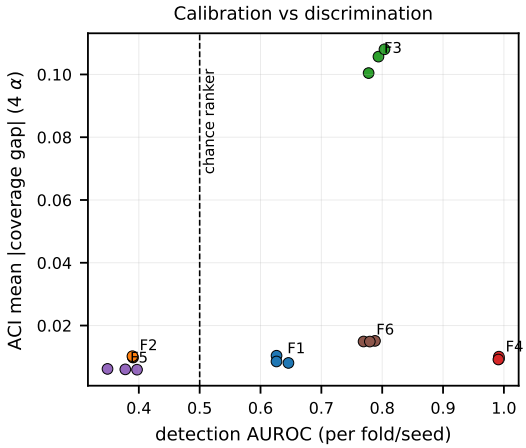


Fig. 4. Calibration versus discrimination: ACI mean coverage gap (four nominal levels) against AUROC for every fold and seed. Below-chance folds (F2, F5) remain coverage-controllable, but their alarms carry no detection utility—an explicit caveat, not a success.

H. Calibration Conclusions Are Not Encoder-Specific

To test whether the calibration findings depend on our neural encoder, we repeated the comparison on folds F1/F4/F6 with two non-neural score producers: PCA reconstruction (32 components on flattened windows) and a Mahalanobis distance on 16-dimensional band-energy features. Table VI reports mean coverage gaps across the four nominal levels. Static calibration fails for these producers as well—including both failure directions *within* the feature-based producer (F4 over-alarms at coverage 0.06; F6 never alarms at 0.000)—while ACI restores gaps to at most 0.011 in all six producer–fold combinations, even when the producer's ranking is below chance (F1 Mahalanobis, AUROC 0.47). The calibration layer is agnostic to score provenance, and the decoupling caveat of Section V-G applies equally to classical detectors.

We further evaluate a *zero-shot time-series foundation model* as an external score producer: Chronos-bolt-small [5], scored by its 32-step forecast residuals (context 224; median quantile; 4,000 stratified evaluation windows per fold, same splits and injection protocol). Table V reports the outcome. The pattern replicates in full: static calibration fails on every fold and in both directions (mean gaps 0.066–0.571); ACI restores gaps to 0.009–0.048; NA-ACI improves further where the budget defect bites (F3: $0.048 \to 0.035$, F6: $0.028 \to 0.014$; F4 range-F_1 0.27 $\to$ 0.39 with κ=3). Two findings stand out. First, Chronos-bolt ranks anomalies *below chance on the same two folds* as our encoder (F2 AUROC 0.34, F5 0.32), which strongly supports reading the inversion as a dataset/label property rather than an encoder artifact. Second, our 1.3M-parameter reconstruction encoder matches or exceeds the zero-shot foundation model on AUROC in all six folds (e.g., F3 0.79 vs. 0.65, F4 0.99 vs. 0.94), consistent with recent evidence that generic foundation models are not automatically superior on shifted industrial series [7], [8].

TABLE V
ZERO-SHOT CHRONOS-BOLT-SMALL EXTERNAL SCORE PRODUCER (FORECAST-RESIDUAL SCORES; 4,000 STRATIFIED WINDOWS PER FOLD). STATIC FAILS BOTH WAYS; ACI RESTORES COVERAGE; NA-ACI TRACKS THE BUDGET DEFECT. AUROC BELOW 0.5 ON F2/F5 MATCHES OUR ENCODER—A DATA/LABEL PROPERTY, NOT AN ENCODER ARTIFACT.

Fold	AUROC	Static gap	ACI gap	NA-ACI gap	R-F_1 ACI	R-F_1 N
F1	0.63	0.124	0.032	0.037 (κ=5)	0.09	0.10
F2	0.34	0.571	0.015	0.041 (κ=5)	0.13	0.13
F3	0.65	0.139	0.048	0.035 (κ=5)	0.30	0.30
F4	0.94	0.066	0.009	0.006 (κ=5)	0.27	0.39
F5	0.32	0.324	0.012	0.012 (κ=5)	0.25	0.28
F6	0.74	0.098	0.028	0.014 (κ=5)	0.37	0.40

TABLE VI
MEAN COVERAGE GAP OVER FOUR NOMINAL LEVELS FOR NON-NEURAL SCORE PRODUCERS (STATIC VS. ACI). ACI RESTORES COVERAGE IN EVERY COMBINATION; STATIC FAILS, INCLUDING IN BOTH DIRECTIONS WITHIN THE MAHALANOBIS PRODUCER.

Producer	F1	F4	F6
PCA reconstruction (static / ACI)	0.154 / **0.002**	0.153 / **0.004**	0.163 / **0.**
Mahalanobis band (static / ACI)	0.546 / **0.004**	0.798 / **0.002**	0.838 / **0.**

I. Comparison with Rival Cross-Domain Detectors (Mechanism-Level)

Because the official codebases of CICADA [3] and DACAD [4] target different benchmarks, we reimplement their core mechanisms under our six-fold protocol and label the comparison accordingly: a *DACAD-style* scorer (anomaly injection + discriminative head + source–target moment alignment) and a *CICADA-style* scorer (four reconstruction experts over source groups with divergence-based expert weighting). Table VII reports ranking quality and calibration. Three observations. First, the calibration result is universal: static thresholds fail for both rivals on every fold (mean gaps 0.037–0.499, including DACAD-style F2 at 0.037 and CICADA-style F2 at 0.499), ACI restores them to 0.001–0.101, and NA-ACI helps where the budget defect bites (CICADA-style F3 gap $0.101 \to 0.059$; F4 range-F_1 0.24 $\to$ 0.72). Second, ranking quality of the rivals is unstable across domains—DACAD-style ranges from AUROC 0.26 (F4) to 0.85 (F3), and CICADA-style from 0.32 (F5) to 0.99 (F4)—so neither is a reliable default. Third, no scorer dominates: our plain encoder matches the best rival on F1/F4/F6 and trails on F2/F3/F5, while every scorer's ranking swings across folds (ours 0.37–0.99, DACAD-style 0.26–0.85, CICADA-style 0.32–0.99)—supporting calibration as the first-order problem for deployment, with ranking left fold-dependent.

J. Runtime, Memory, and Negative Controls

The calibration layer is cheap: across all folds, training takes $\sim$70 s, target scoring at most 0.41 s, and one ACI update costs 0.6–0.8 μs, with 26 MB peak GPU memory (Table VIII). The thresholding mechanisms differ negligibly in cost. For completeness we record the negative controls from the predecessor study: spectral-invariance, relation-consistency, and band-saliency losses, low-rank test-time adaptation, and divergence-

TABLE VII
MECHANISM-LEVEL RIVAL COMPARISON (6 FOLDS). AUROC PER SCORER; MEAN COVERAGE GAP STATIC → ACI; RANGE-F_1 (MEAN OVER LEVELS) FOR ACI AND NA-ACI. STATIC FAILS FOR BOTH RIVALS; ACI RESTORES; RIVAL RANKINGS ARE UNSTABLE ACROSS FOLDS. BEST PER-ROW AUROC IN BOLD.

	AUROC			Gap (static→ACI)	
Fold	Ours	DACAD*	CICADA*	DACAD*	CICADA*
F1	**0.63**	0.60	0.62	.065→.003	.075→.006
F2	0.39	**0.49**	0.35	.037→.003	.499→.010
F3	0.79	**0.85**	0.84	.155→.093	.155→.101
F4	0.99	0.26	**0.99**	.064→.002	.158→.006
F5	0.37	**0.59**	0.32	.076→.002	.238→.004
F6	0.78	0.53	**0.78**	.143→.001	.133→.011

TABLE VIII
PER-RUN COST (ONE FOLD, ONE SEED; RTX 3060 TI). ACI UPDATE LATENCY MEASURED OVER ALL EVALUATION WINDOWS AND NOMINAL LEVELS.

	Train (s)	Score (s)	Update (μs)	Peak mem (MB)	Params
Value	∼70	≤0.41	0.6–0.8	26	∼1.3 M

based source gating each changed range-F_1 by at most 2.2% across four folds, and encoder identity (plain vs. regularized vs. gated) moves calibration coverage by at most 0.002—which is why this paper studies the calibration layer rather than detector architecture.

VI. DISCUSSION AND DEPLOYMENT GUIDELINE

A. What We Would Tell a Practitioner

Four rules follow directly from the study. (1) *Never ship a source-calibrated static threshold for a new domain.* Across four domain families the static threshold failed on six of six folds and in both directions; the failure is not a tuning problem but a distributional one. (2) *Wrap the score with an online ACI layer and monitor its trace; use NA-ACI when anomalies may be abundant.* On five folds the layer recovered coverage within 0.015 of nominal at negligible cost (0.6–0.8 μs per decision); on anomaly-rich streams, NA-ACI keeps the budget on the normal regime and improves detection sharply at matched FPR, at the price of one interpretable parameter κ whose exclusion rate is monitorable online. (3) *Report coverage and alarm rate separately, and know which one is guaranteed.* ACI controls the long-run alarm rate; normal-window FPR control follows when anomalies are rare relative to α, and NA-ACI narrows the gap otherwise. (4) *Audit the ranker before trusting the calibration.* Two folds here rank anomalies below chance; their coverage numbers look excellent and their alarms are worthless. A feature-baseline AUROC check (RMS or spectral energy) costs nothing and exposes this.

B. Limitations

The benchmark has six folds over four domain families, with single-dataset wind and one fab dataset; "cross-domain" here means cross-population within industrial monitoring, not six separate industries. Pseudo-labels are unavoidable for several datasets, and PHM16's q_{10} removal-rate protocol and the PVDAQ synthetic injection are the weakest links; both are documented and their sensitivity is reported (contamination sweep, diagnosis of below-chance folds). The encoder family is deliberately simple and uniform; we compare against mechanism-level reimplementations of the closest rivals (CICADA-, DACAD-style, Section V-I) and against classical and foundation-model score producers (PCA, band-energy, Mahalanobis, Chronos-bolt-small), while official-code comparisons remain a revision item. Finally, the conditional-coverage goal remains open: Mondrian and divergence-shifted variants did not deliver it, and NA-ACI narrows but does not close the conditional gap, so we claim no more than marginal coverage control with an improved budget allocation.

C. Future Work

Three directions follow. First, conditional coverage beyond marginal control: NA-ACI fixes the allocation of the alarm budget but does not provide group-conditional guarantees, and the assumption behind its exclusion rule (anomaly-pure tail) deserves theory that links measurable diagnostics—exclusion rate, score-distribution drift—to coverage bounds. Second, extending the benchmark with more domain families and externally produced scores (foundation models, domain-adaptation detectors) to test whether the calibration conclusions and NA-ACI gains transfer to other score producers. Third, a theory-side account of interval inflation: inflation spans two orders of magnitude across our folds and we can describe it but not yet predict it.

VII. CONCLUSION

Deployed anomaly detectors are thresholds, not scores, and a threshold calibrated on one domain breaks on the next. Across six leave-one-domain-out folds over four domain families of industrial time series, static source-calibrated thresholds failed in both directions—over-alarming to 0.18 coverage on wind data and under-alarming to 0.999 elsewhere—while an online adaptive-conformal layer restored normal-window coverage to within 0.015 of nominal on five folds at 0.6–0.8 μs per decision. Target-aware refinements (divergence-shifted, conditional) brought no gain, divergence did not predict the $0.5\times$–$19\times$ transfer cost, and on two folds whose scores rank anomalies below chance, calibration controlled coverage but could not make the alarms useful. We release the benchmark, protocols, and code, and we state the negative results plainly: deploy calibration, audit the ranker first, and never trust a source threshold on a new domain.

REFERENCES

[1] I. Gibbs and E. Candès, "Adaptive conformal inference under distribution shift," *arXiv preprint arXiv:2106.00170*, 2021.

[2] P. Yan, A. Abdulkadir, P.-P. Luley, M. Rosenthal, G. A. Schatte, B. F. Grewe, and T. Stadelmann, "A comprehensive survey of deep transfer learning for anomaly detection in industrial time series: Methods, applications, and directions," *arXiv preprint arXiv:2307.05638*, 2023.

[3] T. Lan, Y. Gao, Y. Lu, and C. Zhang, "Cicada: Cross-domain interpretable coding for anomaly detection and adaptation in multivariate time series," *arXiv preprint arXiv:2505.00415*, 2025.

[4] Z. Z. Darban, Y. Yang, G. I. Webb, C. C. Aggarwal, Q. Wen, S. Pan, and M. Salehi, "Dacad: Domain adaptation contrastive learning for anomaly detection in multivariate time series," *arXiv preprint arXiv:2404.11269*, 2024.

[5] A. F. Ansari, L. Stella, C. Turkmen, X. Zhang, P. Mercado, H. Shen, O. Shchur, S. S. Rangapuram, S. P. Arango, S. Kapoor, J. Zschiegner, D. C. Maddix, H. Wang, M. W. Mahoney, K. Torkkola, A. G. Wilson, M. Bohlke-Schneider, and Y. Wang, "Chronos: Learning the language of time series," *arXiv preprint arXiv:2403.07815*, 2024.

[6] Y. Nie, N. H. Nguyen, P. Sinthong, and J. Kalagnanam, "A time series is worth 64 words: Long-term forecasting with transformers," *arXiv preprint arXiv:2211.14730*, 2022.

[7] T. Wang, S. Ennadir, J. Pertoft, G. Z. Gandler, L. Cao, Z. Senane, S. Katsarou, S. Asadi, A. Karlsson, and O. Smirnov, "Frequency matters: When time series foundation models fail under spectral shift," *arXiv preprint arXiv:2511.05619*, 2025.

[8] N. Karaouli, D. Coquenet, E. Fromont, M. Mermillod, and M. Reyboz, "How foundational are foundation models for time series forecasting?" *arXiv preprint arXiv:2510.00742*, 2025.

[9] J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani, and L. Wasserman, "Distribution-free predictive inference for regression," *arXiv preprint arXiv:1604.04173*, 2016.

[10] M. Zaffran, A. Dieuleveut, O. Féron, Y. Goude, and J. Josse, "Adaptive conformal predictions for time series," *arXiv preprint arXiv:2202.07282*, 2022.

[11] A. N. Angelopoulos, E. J. Candes, and R. J. Tibshirani, "Conformal pid control for time series prediction," *arXiv preprint arXiv:2307.16895*, 2023.

[12] A. Siffer, P.-A. Fouque, A. Termier, and C. Largou"et, "Anomaly detection in streams with extreme value theory," in *Proc. ACM SIGKDD*, 2017, pp. 1067–1075.

[13] H. R. Medeiros, H. Sharifi-Noghabi, G. L. Oliveira, and S. Irandoust, "Accurate parameter-efficient test-time adaptation for time series forecasting," *arXiv preprint arXiv:2506.23424*, 2025.

[14] PHM Society, "Phm society 2016 data challenge: Cmp semiconductor dataset," Data challenge dataset; official page: https://phmsociety.org/conference/annual-conference-of-the-phm-society/annual-conference-of-the-prognostics-and-health-management-society-2016/phm-data-challenge-4/ (data links defunct; public mirror used, cross-verified), 2016.

[15] C. Gück, C. M. A. Roelofs, and S. Faulstich, "Care to compare: A real-world dataset for anomaly detection in wind turbine data," *arXiv preprint arXiv:2404.10320*, 2024.

[16] NREL, "Pvdaq: Photovoltaic data acquisition benchmark, open energy data initiative," https://oedi-data-lake.s3.amazonaws.com/pvdaq/, 2023.

[17] KAt DataCenter, Paderborn University, "Paderborn bearing run-to-failure dataset under time-varying conditions," Bearing DataCenter, https://mb.uni-paderborn.de/kat/forschung/bearing-datacenter, 2025, original release metadata not stated on the public DataCenter page; subset obtained via a public mirror (2026-05). Verify dataset citation with KAt before camera-ready.

[18] B. Wang, Y. Lei, N. Li, and N. Li, "A hybrid prognostics approach for estimating remaining useful life of rolling element bearings," *IEEE Transactions on Reliability*, vol. 69, no. 1, pp. 401–412, 2020.